NeuroPathX

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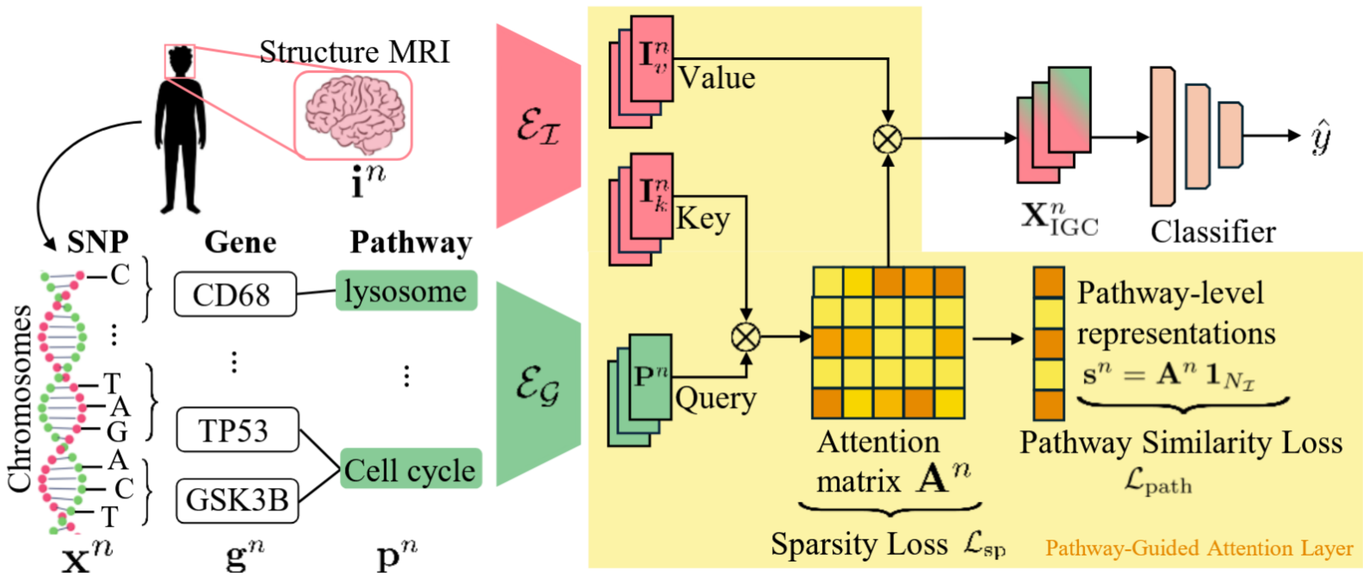
Motivation

- Autism spectrum disorder (ASD) and Alzheimer's disease (AD) arise from complex interactions between **genetics** and **brain structure** — yet most methods treat them separately.
- Genetic variants (SNPs) can be aggregated into biologically meaningful **pathway scores**, linking molecular signals to known cellular functions.
- Structural MRI captures **macroscopic brain changes** associated with neurodegeneration and neurodevelopmental differences.
- **NeuroPathX** jointly models both modalities via a biologically-guided attention mechanism, enabling **interpretable classification** and discovery of pathway–brain region interactions.

Pipeline



Genetic Representation

SNPs $x^n \in \mathbb{R}^{N_G}$, weighted by GWAS β_j

$$g_i^n = \sum_{j \in \Omega_i} \beta_j x_j^n$$

Genes SNPs within ±50kb window

$$p_k^n = \sum_{l \in \Gamma_k} g_l$$

KEGG Pathways $p^n \in \mathbb{R}^{177}$

MRI Representation

Structural MRI (T1-weighted)

Parcellation Brainnetome 210 ROIs (ASD) Desikan-Killiany 68 ROIs (AD)

FreeSurfer: volume, surface area, thickness mean, thickness std ($d = 4$/ROI)

Output $i^n \in \mathbb{R}^{4 \times N_I}$

Loss 1 — Classification

Weighted Cross-Entropy

Handles class imbalance between patients and controls.

$$\mathcal{L}_{\text{CE}} = \sum_n \left[\delta\, y^n \log \hat{y}^n + (1-\delta)(1-y^n) \log(1-\hat{y}^n) \right]$$

δ controls class weighting; y^n true label; $\hat{y}^n$ predicted.

Loss 2 — Sparsity

Attention Sparsity (KL divergence)

Forces attention on the most salient pathway–brain interactions.

$$\mathcal{L}_{\text{sp}} = \sum_i \sum_j \text{KL}\big(\text{Ber}(q) \,\|\, \text{Ber}(A_{ij}^n)\big)$$

Target sparsity $q = 10^{-2}$. Applied to 2 random subjects per batch.

Loss 3 — Pathway Similarity

Cross-Cohort Consistency (ℓ_1)

Encourages consistent pathway patterns across patients and controls.

$$\mathcal{L}_{\text{path}} = \frac{1}{M} \sum_{m=1}^{M} \left\| s_{\text{NC}}^{(m)} - s_{\text{PAT}}^{(m)} \right\|_1$$

$s^n = A^n \mathbf{1}_{N_I}$ (pathway influence); $M = 10$ pairs per epoch.

Results

ACE — ASD
$N = 165$ subjects
Brainnetome atlas (210 ROIs)
10-fold cross-validation

ADNI — Alzheimer's
$N = 438$ subjects
Desikan-Killiany atlas (68 ROIs)
10-fold cross-validation

NeuroPathX achieves best **accuracy** and **specificity** on both cohorts.
Ablation confirms each regularization loss contributes independently to performance.

Limitations & Future Work

Small datasets.
ACE (N=165) and ADNI (N=438) limit the stability of learned attention patterns.

Attention ≠ causation.
Weights reflect correlation, not causal biological relationships.

Fixed biological priors.
The ±50kb window and KEGG pathways may miss context-dependent effects.

LaTeX TikZposter